THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · HARD LEVEL FULL-YEAR PAPER SET 2

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. Two dice are rolled together. The probability that the difference of the numbers is 2 is: 1

(a) $\frac{2}{9}$ **(b)** $\frac{1}{3}$ **(c)** $\frac{7}{9}$ **(d)** $\frac{1}{5}$

2. The sum of the first n terms of a sequence is $S_n = 2n^2 + 3n$. Its 9th term is: 1

(a) 35 **(b)** 41 **(c)** 37 **(d)** 39

3. For the expression $n^2 - n + 11$, the smallest natural number n for which the value is NOT prime is: 1

(a) 11 **(b)** 13 **(c)** 12 **(d)** 10

4. In what ratio must a 25% solution and a 41% solution be mixed to obtain a 34% solution? 1

(a) $8 : 9$ **(b)** $7 : 9$ **(c)** $9 : 7$ **(d)** $7 : 10$

5. Using Euclid's subtraction algorithm for gcd$(1344, 872)$, the number of subtraction steps needed is: 1

(a) 15 **(b)** 16 **(c)** 13 **(d)** 14

6. The sides of a rectangle are in the ratio $3 : 4$ and its diagonal is 25 cm. Its perimeter is: 1

(a) 75 cm **(b)** 70 cm **(c)** 105 cm **(d)** 95 cm

7. A taxi fare is ₹243 for 5 km and ₹425 for 12 km (fare is a linear function of distance). The fare for 38 km (in ₹) is: 1

(a) 1127 **(b)** 1101 **(c)** 1214 **(d)** 1075

8. Take $\pi = \frac{22}{7}$. A toy is a cone of height 6 cm placed on a hemisphere, both of radius 21 cm. Its total volume is: 1

(a) 24394 cm^3 **(b)** 22176 cm^3 **(c)** 26613 cm^3 **(d)** 19958 cm^3

9. A circle has diameter with endpoints $(1, -2)$ and $(19, 78)$. Its radius is: 1

(a) 40 **(b)** 43 **(c)** 42 **(d)** 41

10. If $p(x) = 2x + 3$, then $p(p(4)) =$ 1

(a) 26 **(b)** 24 **(c)** 25 **(d)** 27

11. The smallest natural number n for which $\frac{n}{120}$ has a terminating decimal expansion is: 1

(a) 5 **(b)** 2 **(c)** 4 **(d)** 3

12. If $x + \frac{1}{x} = 3$, then $x^3 + \frac{1}{x^3} =$ 1

(a) 20 **(b)** 18 **(c)** 19 **(d)** 17

13. Two circles of radii 13 cm and 15 cm intersect in a common chord of length 24 cm and their centres lie on opposite sides of it. The distance between the centres (in cm) is: 1

(a) 14 **(b)** 15 **(c)** 13 **(d)** 16

14. Two concentric circles have radii 14 cm and 21 cm (take $\pi = \frac{22}{7}$). The area of the ring between them is (cm^2): 1

(a) 772 **(b)** 769 **(c)** 771 **(d)** 770

15. Three coins are tossed. The probability of getting at least one head is: 1

(a) $\frac{7}{9}$ **(b)** $\frac{1}{8}$ **(c)** $\frac{7}{8}$ **(d)** $\frac{3}{4}$

16. The sum of all integers from 1 to 100 that are divisible by 6 is: 1

(a) 810 **(b)** 826 **(c)** 816 **(d)** 822

17. **Assertion (A):** In a circle of radius 5 cm, a chord at a distance of 3 cm from the centre has length 8 cm. 1

Reason (R): The perpendicular from the centre of a circle to a chord bisects the chord.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

18. **Assertion (A):** The area of a triangle with sides 13 cm, 14 cm and 15 cm is 84 cm^2. 1

Reason (R): By Heron's formula the area is $\sqrt{s(s-a)(s-b)(s-c)}$, where s is the semi-perimeter.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

19. **Assertion (A):** When a fair die is rolled, the probability of getting a prime number is $\frac{1}{2}$. 1

Reason (R): The prime numbers on a die are 2, 3 and 5, which are 3 of the 6 equally likely outcomes.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

20. **Assertion (A):** The sequence $2, 5, 8, 11, \ldots$ is an arithmetic progression. 1

Reason (R): The difference between consecutive terms is the constant 3.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 6 × 2 = 12

Answer in about 20–30 words each.

21. Two dice are rolled together. Find the probability that the product of the numbers is 12. 2

22. The sum of the first 15 terms of an AP is 255 and the first term is 3. Find d and the 15th term. 2

23. Use the converse of the Baudhāyana–Pythagoras theorem: for what value of x is a triangle with sides $20, 21, x$ (x the longest) right-angled? 2

24. A chemist has a 16% acid solution and a 64% acid solution. In what ratio must they be mixed to get a 36% solution? 2

25. Find the gcd of 380, 520 and 152 by applying Euclid's subtraction algorithm twice: first to two numbers and then to the result and the third. 2

26. A rhombus has side 51 cm and one diagonal 48 cm. Find the other diagonal, its area and the distance between a pair of opposite sides (height), correct to two decimal places. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

27. Prove that the sum of the squares of two consecutive integers is odd. Verify it for 8 and 9. 3

28. In a class, 40 students have an average of 49 marks and the remaining students have an average of 90 marks. The class average is 70. Find the number of remaining students. 3

29. Trace: count = 0; for i = 1 to 11: for j = i to 11: count = count + 1. Find the final count and explain the pattern. 3

30. The diagonals of a parallelogram ABCD meet at O. $OA = (5x + 3)$ cm and $OC = (x + 147)$ cm. Find x and the length of AC. 3

31. A line cuts the x-axis at $(18, 0)$ and the y-axis at $(0, 17)$. Find its equation, its slope and the area of the triangle it makes with the axes. 3

OR

A solid sphere of radius 12 cm is melted and recast into a solid cylinder of radius 3 cm. Find the height of the cylinder and its curved surface area (Take $\pi = \frac{22}{7}$). 3

32. Three vertices of a rectangle $ABCD$ are $A(0, 0)$, $B(9, 12)$ and $D(-16, 12)$. Find C and the diagonal AC. 3

OR

A linear polynomial has $p(1) = 12$ and $p(3) = 20$. Find $p(x)$, its zero and $p(6)$. 3

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. **Case study.** Mrs Gupta writes the fractions $\frac{7}{16}$, $\frac{5}{12}$ and $\frac{11}{25}$ on the board and asks the class to find their decimal expansions. 5

(i) Write $\frac{7}{16}$ as a decimal.

(ii) Write $\frac{11}{25}$ as a decimal.

(iii) Express the denominator 12 in prime factors.

(iv) Which of the three fractions have terminating decimal expansions? Give the reason.

34. **Case study.** Sunita, a Class 9 student in Lucknow, calculates squares and products quickly using algebraic identities instead of long multiplication. 5
(i) Find 103^2.
(ii) Find 98^2.
(iii) Find 102×98.
(iv) Find 997×1003 using an identity and name the identity.

35. The midpoints of the sides of a rectangle 14 cm by 48 cm are joined in order. Show that the figure formed is a rhombus and find its perimeter and area. 5

36. A line passes through $(-1, 1)$ and $(5, 7)$. Find its equation, then find k if the point $(k, 9)$ lies on it, and the x-intercept. 5

OR

Take $\pi = \frac{22}{7}$. An ice-cream cone of radius 42 cm and height 56 cm is completely filled and topped with a hemisphere of ice-cream of the same radius. Find the total volume of ice-cream and the volume needed for 27 such cones. 5

37. The point $P(x, 0)$ is equidistant from $A(3, 4)$ and $B(9, -4)$. Find P and the distance PA. 5

OR

A line passes through $(1, 15)$ and $(4, 24)$. Find its slope, equation, intercepts and the area of the triangle it forms with the axes. 5

38. Simplify $(\sqrt{7} + \sqrt{5})^2 + (\sqrt{7} - \sqrt{5})^2$ and $(\sqrt{7} + \sqrt{5})(\sqrt{7} - \sqrt{5})$. Which are rational? 5

OR

If $x + y = 10$ and $x^2 + y^2 = 58$, find xy, $(x - y)^2$ and $x^2 - y^2$ given that $x > y$. 5

END OF QUESTION PAPER **Total: 80 Marks**

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